PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE
CALCIUM CHLORIDE CONCENTRATION POST
氯化钙浓缩岗位

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the preparation of the calcium chloride concentration post under complete closure.

此程序涵盖操作员对完全关闭下的氯化钙浓缩岗位进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that preparation of the calcium chloride concentration post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the preparation of the calcium chloride concentration post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对氯化钙浓缩岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定氯化钙浓缩岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures.
  MTI-CP-SOP岗位操作规程
- Concentration Tower(4307-TOW-001).
  逆喷浓缩塔(4307-TOW-001)
- Draft Fan(4307-FAN-001).
  引风机(4307-FAN-001)
- Dilute CaCl2 Solution Storage Tank(4307-TNK-002/003).
  稀氯化钙溶液储槽(4307-TNK-002/003)
- Dilute CaCl2 Solution Pump(4307-PSO-001/002).
  稀氯化钙溶液泵(4307-PSO-001/002)
- Evaporative Circulating Pump(4307-PSO-005/006).
  蒸发循环泵(4307-PSO-005/006)
- Filter Press Feed Pump(4307-PSO-007/008).
  烟尘压滤泵(4307-PSO-007/008)
- Dust Filter Press(4307-FIL-001).
  烟尘压滤机(4307-FIL-001)

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4. PPE 个人保护装备





5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Poisoning; 3. Electric shock; 4. Slippery conditions and a trip to the danger; 5. Asphyxia; 6. Object falling from high altitudes; 7. Empyrosis; 8. Vehicle collision crush; 9. Liquid sputtering; 10. Liquid corrosion; 11. Noise. 1. 机械伤害; 2. 中毒; 3. 触电; 4. 湿滑的条件，绊倒危险; 5. 窒息; 6. 高空坠物; 7. 烫伤; 8. 车辆碰撞碾压; 9. 液体喷溅; 10. 液体腐蚀; 11. 噪音	1. Install a protective cover on the rotating part of the equipment; 2. Wear good labor protection goods, Found smoke or liquid running phenomenon immediately treated and reported; 3. Do not contact the electrical switch with wet hands during operation, Regular inspection of the equipment leakage protection device is complete, effective; 4. Please hold the handrail up and down the stairs; 5. Clear clearance or maintenance operations; 6. Wear labor protection goods and check the safety of stacking items on the operation platform; 7. Wear good labor protection goods; 8. Walk along the prescribed route, and stay away from the dust slag transport vehicles; 9. Wear protective masks and gloves, as well as other protective products as required; 10. Wear appropriate labor protection supplies, once the skin into contact with a corrosive solution, immediately rinse with a lot of water; 11. Wear the earplugs to protect your hearing

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Description of Hazard / Environmental Issue 危害	Control Strategy 控制
	1. 设备的转动部位安装防护罩； 2. 穿戴好劳保护品，发现烟气或液体跑冒现象立即处理并汇报； 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效； 4. 上下楼梯请扶好扶手； 5. 清仓或维修作业执行审批程序； 6. 穿戴好劳保护品，检查各平台的物品堆放安全性; 7. 穿戴好劳保护品； 8. 按照规定的路线行走，远离烟尘渣运输车辆； 9. 按要求戴好防护面具和防护手套，以及其它劳保护品； 10. 穿戴适宜的劳保用品，一旦皮肤接触到有腐蚀的溶液，立即用大量清水冲洗； 11. 佩戴好耳塞，保护听力

6. Equipment 工具/ 设备

4307-TNK-001	Overhead Tank高位清液槽
4307-PSO-009	Submerged Pump液下泵
4307-CHY-001	Chimney烟囱
4307-CAT-201	Compressed Air Tank压缩空气储气罐
4307-CHE-001	Smoke Filter Press Chute烟尘压滤机溜槽
	Clean the site tools清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions.

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№	Stage	Key Points
		1. 确认经过岗前技术培训； 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患； 3. 检查并打扫设备、现场卫生； 4. 与前后工序取得联系，并发出明确指令
2	Inspect Concentration Tower (4307-TOW-001) 检查逆喷浓缩塔(4307-TOW-001)	1. Remove all sundries in the reverse spray concentration tower and other equipment; 2. Check whether the instruments are complete, and whether the liquid level meter, pressure and thermocouple display are normal; 3. Check whether the liquid circulation pipeline, gas pipeline and valve are in good condition; 4. Open the new water brake valve (60GA) and fill the high clearance tank (4307-TNK-001) with new water; 5. Open the high liquid tank (4307-TNK-001) to drain two butterfly valves (100BF) and fill the lower part of the reverse concentration tower with new water; 6. Open the evaporation cycle pump (4307-PSL-005/006), Observe the spray and washing condition; 7. Check whether the overflow weir and nozzle assembly of the spray tower are normal and there is no blockage; 8. Check whether the circulating liquid inlet, circulating liquid outlet, backup port, gas inlet and outlet, manhole, and hand hole are normal, whether the connection bolts are fastened, leakage or leakage, and check whether the tower body is leakage or leakage; 9. Empty the inspection water in the reverse concentration tower. 1. 清除逆喷浓缩塔等设备内的一切杂物； 2. 检查各处仪表是否齐全，液位计、压力和热电偶显示是否正常； 3. 检查各液体管线、气体管路、阀门是否完好； 4. 打开新水闸阀(60GA)，将高位清液槽(4307-TNK-001)注满新水； 5. 打开高位清液槽(4307-TNK-001)放水两个蝶阀(100BF)，将逆喷浓缩塔下部充满新水； 6. 开启蒸发循环泵(4307-PSL-005/006)，观察喷淋洗涤情况； 7. 检查逆喷浓缩塔溢流堰、喷嘴组件是否正常，有无堵

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№	Stage	Key Points
		塞现象； 8. 检查逆喷浓缩塔循环液进口、循环液出口、备用口、气体进出口、人孔、手孔等是否正常，连接螺栓是否紧固，是否漏液或漏气；检查塔体是否漏液或漏气； 9. 放空逆喷浓缩塔内的检查用水。
3	Inspect Draft Fan (4307-FAN-001) 检查引风机 (4307-FAN-001)	1. Manual turning 2~3 turn, flexible rotation, no scraping, stuck and other abnormal phenomena; 2. The fan air outlet valve is normal, and the fan inlet valve is closed; 3. Whether the coupling concentricity meets the requirements; 4. Check whether the lubricating oil level of the fan bearing box is normal; 5. Close the power switch, first click (press the operation button and quickly press the stop button) to confirm that the steering is correct, no noise, no vibration and other abnormal conditions; 6. After the fan is started and running, observe the change of the fan sound, check the temperature and noise of the fan bearing and the motor bearing, and check the fan vibration (including the impeller). 7. 手动盘车2 ~ 3转，转动灵活，不得有刮蹭、卡住等异常现象； 8. 风机出风阀门正常，风机进口阀门关闭； 9. 联轴器同心度是否符合要求； 10.检查风机轴承箱润滑油位是否正常； 11.合上电源开关，先点动(按运转按钮后快速按停止按钮)确认转向正确、无杂音、无振动等异常情况； 12.风机启动运行后，观察风机声音的变化，检查风机轴承、电机轴承的温度和噪声，检查风机振动(包括叶轮)
4	Inspect Dilute CaCl2 Solution Storage Tank(4307-TNK-002/003) 检查稀氯化钙溶液储槽(4307-TNK-002/003)	1. Check whether the inlet and outlet pipelines and auxiliary pipelines, whether the installation of flange and valve meet the requirements, and whether the storage tank and dilute acid tank leaks liquid; 2. Check the level control assembly is normal. 1. 检查出入口管线及附属管线，法兰、阀门安装是否符合要求，循环槽和稀酸槽是否漏液； 2. 检查液位控制组件是否正常

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№	Stage	Key Points
5	Inspect Dilute CaCl2 Solution Pump (4307-PSO-001/002), Evaporative Circulating Pump (4307-PSO-005/006), Filter Press Feed Pump (4307-PSO-007/008) 检查稀氯化钙溶液泵(4307-PSO-001/002)，蒸发循环泵(4307-PSO-005/006)，烟尘压滤泵(4307-PSO-007/008)	1. Check whether the pump inlet pipeline and auxiliary pipeline, flange and valve installation meet the requirements, whether the anchor bolts and ground wire are good, and whether the coupling is installed; 2. Turn the pump shaft 3 times, should be flexible without block; 3. Check whether the lubricating oil level is normal. The oil level should be between 1/2 and 2/3 of the oil lens; 4. Check the sealing condition of the pump is normal; 5. Point to move to confirm the correct steering; 6. Check whether the voltage meter, signal lamp and other meters of the pump control cabinet are normal; 7. Check whether the pump unit has air, if so, should be excluded; 8. Check whether the pump head is under the tank of dilute calcium chloride solution (4307-TNK-003) or the reverse concentration tower (4307-TOW-001). 1. 检查泵出入口管线及附属管线，法兰、阀门安装是否符合要求，地脚螺栓及地线是否良好，联轴器及防护罩是否装好； 2. 手转动泵轴3圈，应灵活无阻滞； 3. 检查润滑油油位是否正常，油位应在油镜的1/2 ~ 2/3 之间； 4. 检查泵的密封状况是否正常； 5. 点动确认转向正确； 6. 检查泵控制柜电压表、信号灯等仪表指示是否正常； 7. 检查泵机组是否有空气，如有应予以排除； 8. 检查泵头是否处于稀氯化钙溶液储槽(4307-TNK-003)或逆喷浓缩塔(4307-TOW-001)液位下
6	Inspect Dust Filter Press (4307-FIL-001) 检查烟尘压滤机(4307-FIL-001)	1. Check whether the number of filter plates is sufficient, whether there is damage, whether the filter plate is clean, and whether the placement meets the requirements; 2. Check whether the filter cloth is folded, whether it is damaged, and whether the filter performance is good; 3. Check whether the pressure gauge of the mine pipeline is normal; 4. Check whether the connection is close and whether there is leakage; 5. Check whether the oil of the filter press is sufficient and whether the oil level meets the requirements (1 / 2~1 / 3); 6. Check whether the hydraulic oil top device is normal; 7. Check whether the emergency stop device is normal;

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		8. Check whether the pressure gauge of the hydraulic oil is normal and whether the pointer is in the specified interval; 9. Check whether the import and export of the filter press are smooth; 10. Automatic plate drawing device; 11. Whether the water connection plate is flexible and easy to use. 1. 检查滤板数量是否足够，有无破损，滤板是否清洁，安放是否符合要求; 2. 检查滤布是否折叠，有无破损，过滤性能是否良好; 3. 检查给矿管路压力表是否正常; 4. 检查各处联接是否紧密，有无泄露; 5. 检查压滤机油是否足够，油位是否符合要求(1/2 ~ 1/3); 6. 检查液压油顶装置是否正常; 7. 检查急停装置是否正常; 8. 检查液压油的压力表是否正常，指针是否在规定的区间; 9. 检查压滤机各进出口是否畅通; 10. 检查自动拉板装置良好; 11. 接水盘是否灵活好用。
7	Check the other Equipment and Facilities 检查其它设备设施	1. Overhead Tank(4307-TNK-001) a. Check whether the inlet and outlet pipelines and auxiliary pipelines, whether the installation of flange and valve meet the requirements, and whether the groove body is leaking liquid; b. Check the level control assembly is normal. 1. 高位清液槽(4307-TNK-001) a. 检查出入口管线及附属管线，法兰、阀门安装是否符合要求，槽体是否漏液; b. 检查液位控制组件是否正常。 2. Submerged pump(4307-PSO-009) a. Check whether there is debris in the lower pump tank and the pipeline blockage; b. Check whether the fastening screws of the equipment and motor are loose, whether the bearing is intact, and whether the lubricating oil is intact; c. Check whether the electrical part of the equipment

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№	Stage	Key Points
		is in good condition and whether the motor shell is well grounded; d. Check whether the steering of the motor is correct; e. Check whether the pump head is under the ground trough liquid level. 2. 液下泵(4307-PSO-009) a. 检查液下泵槽内有无杂物，管路堵塞情况； b. 检查设备、电机的紧固螺丝是否松动，轴承是否完好，润滑油是否完好； c. 检查设备电气部分是否完好，电机外壳是否接地良好； d. 检查电机的转向是否正确； e. 检查泵头是否处于地槽液位下。 3. Chimney(4307-CHY-001) a. Check whether there is debris at the bottom of the chimney, whether there is leakage and ventilation is unobstructed. 3. 烟囱(4307-CHY-001) a. 检查烟囱底部是否有杂物，是否漏气，通风是否通畅。 4. Compressed Air Tank(4307-CAT-201) a. Check whether the connecting bolts of each flange are loose; b. Check whether all the air pipes and connections are loose and leaky; c. Check whether all the pneumatic components are working normally and flexibly. 4. 压缩空气储气罐(4307-CAT-201) a. 检查各个法兰的连接螺栓是否松动； b. 检查全部空气管道及连接处是否松动漏气； c. 检查全部气动元件工作是否正常、灵活。 5. Smoke Filter Press Chute (4307-CHE-001) a. Check whether the chute is in good condition; b. Check whether the blocking material detection switch of the chute is normal. 5. 烟尘压滤机溜槽(4307-CHE-001) a. 检查溜槽是否完好； b. 检查溜槽的堵料检测开关是否正常。

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		6. Valve a. Check whether the compressed air storage tank (4307-CAT-201) (50BA) two air ball valves (50BA, 32BA) and outlet two stop valves (32GB, 32GB) are normal and flexible; b. Check whether the new water inlet, one gate valve (65GA) of the high clearance tank (4307-TNK-001), and the outlet, four butterfly valves (100BF, 100BF, 100BF, 100BF), Line valve (80BA) coming from the chloride pump (4307-PSO-003/004) are normal and flexible; c. Check whether a butterfly valve (150BF) at the bottom of the dilute calcium chloride solution (4307-TNK-002), and the butterfly valve (150BF) at the bottom of the dilute calcium chloride solution (4307-TNK-003) is normal and flexible; d. Check whether a butterfly valve (150BF) at the bottom of the dilute calcium chloride solution reservoir (4307-TNK-003), a butterfly valve (150BF) at the bottom of the dilute calcium chloride solution reservoir (4307-TNK-002), and a ball valve (200BF) from the lead-zinc filter (4402-FIL-005/006) is normal and flexible; e. Check whether the two butterfly valves (150BF, 150BF) at the inlet of the dilute calcium chloride solution pump (4307-PSO-001/002), and the two butterfly valves (100BF, 100BF) are normal and flexible; f. Check whether the four butterfly valves (100BF, 100BF, 100BF, 100BF) (4307-TOW-001) are normal and flexible; g. Check whether the two butterfly valves (500BF, 500BF) at the entrance of the evaporation circulation pump (4307-PSO-005/006) are normal and flexible; h. Check whether the two butterfly valves (150BF, 150BF, 150 BF) at the inlet of the dust pressure filter pump (4307-PSO-007/008), and the two butterfly valves (125BF, 125BF) are normal and flexible; i. Check whether the filtrate of the dust filter press (4307-FIL-001) flows to a butterfly valve (150BF) in the reverse spray concentration tower (4307-TOW-001) and a butterfly valve (150BF) in the concentrated calcium chloride supernatant reservoir (4307-TNK-004) is normal and flexible; j. Check whether the inlet and outlet butterfly valve of

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№	Stage	Key Points
		the induced draft fan (4307-FAN-001) (2200BF, 2200BF) is normal and flexible. 6. 阀门 a. 检查压缩空气储气罐(4307-CAT-201)一个进出气球阀(50BA)两个出气球阀(50BA,32BA)和出口两个截止阀(32GB,32GB)是否正常、灵活; b. 检查高位清液槽(4307-TNK-001)新水入口一个闸阀(65GA)，出口三个蝶阀(100BF,100BF,100BF)；浓氯化钙清液泵(4307-PSO-003/004)泵过来的液体管路闸阀(80BA)是否正常、灵活; c. 检查稀氯化钙溶液储槽(4307-TNK-002)底部一个蝶阀(150BF)，底部和稀氯化钙溶液储槽(4307-TNK-003)相连的底部一个蝶阀(150BF)是否正常、灵活; d. 检查稀氯化钙溶液储槽(4307-TNK-003)底部一个蝶阀(150BF)，底部和稀氯化钙溶液储槽(4307-TNK-002)相连的底部一个蝶阀(150BF)，铅锌压滤机(4402-FIL-005/006)过来的压滤液一个球阀(200BF)是否正常、灵活; e. 检查稀氯化钙溶液泵(4307-PSO-001/002)的入口两个蝶阀(150BF,150BF)，出口两个蝶阀(100BF,100BF)是否正常、灵活; f. 检查逆喷浓缩塔(4307-TOW-001)的来自高位清液槽(4307-TNK-001)新水入口四个蝶阀(100BF,100BF,100BF,100BF)是否正常、灵活; g. 检查蒸发循环泵(4307-PSO-005/006)的入口两个蝶阀(500BF,500BF)是否正常、灵活; h. 检查烟尘压滤泵(4307-PSO-007/008)的入口两个蝶阀(150BF,150BF)，出口两个蝶阀(125BF,125BF)是否正常、灵活; i. 检查烟尘压滤机(4307-FIL-001)滤液流向逆喷浓缩塔(4307-TOW-001)的一个蝶阀(150BF)，流向浓氯化钙清液储槽(4307-TNK-004)的一个蝶阀(150BF)是否正常、灵活;

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№	Stage	Key Points
		j. 检查引风机(4307-FAN-001)进出口蝶阀(2200BF,2200BF)是否正常、灵活。
8	Inspect Materials Level 检查物料料位	1. Ensure that the gas storage tank (4307-CAT-201) is filled with compressed air and under normal pressure before operation; 2. Check whether the high level liquid clearance tank (4307-TNK-001) and the dilute calcium chloride solution storage tank (4307-TNK-002/003) are filled with new water; 3. Check whether the circulating water at the bottom of the reverse concentration tower (4307-TOW-001) is sufficient; 4. Check whether the material is stored in the gold mud filter press (4402-FIL-012), whether the hydraulic oil is sufficient, and whether there is a filter cake in the feeding chute. 1. 操作前应确保储气罐(4307-CAT-201)充满压缩空气，压力正常； 2. 检查高位清液槽(4307-TNK-001)和稀氯化钙溶液储槽(4307-TNK-002/003)是否充满新水； 3. 检查逆喷浓缩塔(4307-TOW-001)底部循环水是否足够； 4. 检查烟尘压滤机(4307-FIL-001)内是否存料，液压油是否足够，下料溜槽是否存有滤饼
9	Start-Up Of Calcium Chloride Concentration Post 氯化钙浓缩岗位开机	1. No-load start should be guaranteed when starting; 2. The driving sequence is along the material direction, from back to front that is the reverse drive, Each equipment needs to be switched on later;At the same time, Confirm that the process behind preparation of the calcium chloride concentration post has been started and has the feeding conditions: Open the liquid pipeline gate valve (80BA) from the concentrated calcium chloride liquid pump (4307-PSO-003/004), fill the high liquid tank (4307-TNK-001) with calcium chloride solution, Open the two direct butterfly valves (100BF, 100BF) at the outlet, open the four butterfly valves (100BF,100BF,100BF,100BF) from the high spray tank (4307-TOW-001), After the bottom of the reverse spray concentration tower (4307-TOW-001) is filled with calcium chloride solution, the calcium chloride solution overflows into the dilute calcium chloride solution tank (4307-TNK-003), and the dilute

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		calcium chloride solution tank (4307-TNK-003) is filled, Open the bottom two butterfly valves (150BF, 150BF) connected to the dilute calcium chloride solution reservoir (4307-TNK-002), and then fill the dilute calcium chloride solution reservoir (4307-TNK-002) with the calcium chloride solution⟹ In the above process, ensure that the high clearance tank (4307-TNK-001) after the dilute calcium chloride solution tank (4307-TNK-002/003) are filled with calcium chloride solution (4307-TW-001)⟹ Close two direct butterfly valves (100BF, 100BF) at the outlet of the high tank (4307-TNK-001), and open a gate valve (65GA) at the inlet of the high tank (4307-TNK-001) with new water⟹ Open the outlet butterfly valve (2200BF) (4307-FAN-001), start the induced draft fan (4307-FAN-001), and open the imported induced draft fan (4307-FAN-001) (2200BF)⟹Open the outlet of two butterfly valves (100BF, 100BF) of the dilute calcium chloride solution pump (4307-PSO-001/002), start the dilute calcium chloride solution pump (4307-PSO-001/002), and open the two butterfly valves (150BF) (4307-PSO-001/002)⟹ Before the arrival of the hot air of the ring cooler (4305-COO-001), Open the evaporating cycle pump (4307-PSO-005/006), Open the evaporation cycle pump (4307-PSO-005/006) inlet with two butterfly valves (500BF, 500BF)⟹ Start the dust filter press (4307-FIL-001)When the impurities in the calcium chloride solution at the bottom of the reverse spray concentration tower (4307-TOW-001) reaches the required concentration, press the "start" button of the dust filter press (4307-FIL-001), and the oil pump starts working, Then press the "press" button, the piston pushes the pressing plate press, when the pressure force reaches the high pressure, the hydraulic system automatically stop and maintain the pressure, Open the inlet butterfly valve (125BF) of the filter press and one butterfly valve (150BF) of the filter press to the concentrated calcium chloride reservoir (4307-TNK-004) ⟹Open the two butterfly valves (125BF, 125BF) (4307-PSO-007/008), start the dust filter pump (4307-PSO-007/008), and open the two butterfly valves (150BF, 150BF)⟹The slurry is filtered into each filter chamber through the feed hole on the thrust plate, and then filtered under the feed pressure not greater than 0.6MPa to gradually form the filter cake⟹ After the filter cake is filled, close the two butterfly

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№	Stage	Key Points
		valves (150BF-PSO-007/008), stop the dust filter pump (4307-PSO-007/008), close the dust filter pump (4307-PSO-007/008), two butterfly valves (125BF, 125BF) and the filter inlet butterfly valve (125BF)⟹Slowly open the compressed air reservoir (4307-CAT-201), an inlet and outlet globe valve (50BA), two globe valves (50BA, 32BA) and outlet two globe valves (32GB, 32GB), dry the filter cake at a pressure not greater than 0.6MPa. After operation, close the compressed air reservoir (4307-CAT-201), two globe valves (50BA, 30 BA) and outlet two globe valves (32GB, 32GB, 32 GB)⟹ Hold the "release" button on the operation panel of the dust filter press (4307-FIL-001), and the pressure plate will automatically stop after it is returned in place. At the same time, the pull hook will move forward to discharge the reciprocating plate⌋ Wait for notification to open or close a ball valve from the lead-zinc filter press (4402-FIL-005/006) (200BF). 1. 启动时应保证空载启动； 2. 开车顺序为沿物料走向，从后往前开即逆向开车，各设备都需要延时开启；同时，确认氯化钙浓缩岗位后面的工序已经开车并具备给料条件： 打开浓氯化钙清液泵(4307-PSO-003/004)泵过来的液体管路闸阀(80BA)，将高位清液槽(4307-TNK-001)注满氯化钙溶液，打开出口两个直通的蝶阀(100BF,100BF)，打开逆喷浓缩塔(4307-TOW-001)的来自高位清液槽(4307-TNK-001)新水入口四个蝶阀(100BF,100BF,100BF,100BF)，使逆喷浓缩塔(4307-TOW-001)底部充满氯化钙溶液，氯化钙溶液溢流到稀氯化钙溶液储槽(4307-TNK-003)，稀氯化钙溶液储槽(4307-TNK-003)充满后，打开与稀氯化钙溶液储槽(4307-TNK-002)相连的底部两个蝶阀(150BF,150BF)，再使稀氯化钙溶液储槽(4307-TNK-002)充满氯化钙溶液⟹上述过程中，保证在逆喷浓缩塔(4307-TOW-001)和稀氯化钙溶液储槽(4307-TNK-002/003)都充满氯化钙溶液后，高位清液槽(4307-TNK-001)是空槽状态⟹将高位清液槽(4307-TNK-001)出口两个

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		直通的蝶阀(100BF,100BF)关闭，打开高位清液槽(4307-TNK-001)新水入口一个闸阀(65GA)，注满新水⟹打开引风机(4307-FAN-001)出口蝶阀(2200BF)，启动引风机(4307-FAN-001)，打开引风机(4307-FAN-001)进口蝶阀(2200BF)⟹打开稀氯化钙溶液泵(4307-PSO-001/002)出口两个蝶阀(100BF,100BF)，启动稀氯化钙溶液泵(4307-PSO-001/002)，打开稀氯化钙溶液泵(4307-PSO-001/002)入口两个蝶阀(150BF,150BF)⟹在环冷机(4305-COO-001)的二三段热风到达之前，开启蒸发循环泵(4307-PSO-005/006)，打开蒸发循环泵(4307-PSO-005/006)入口两个蝶阀(500BF,500BF)⟹启动烟尘压滤机(4307-FIL-001)⌊待逆喷浓缩塔(4307-TOW-001)底部氯化钙溶液中杂质达到要求的浓度时，按下烟尘压滤机(4307-FIL-001)“启动”按钮，油泵开始工作，然后再按一下“压紧”按钮，活塞推动压紧板压紧，当压紧力到达调定高点压力后，液压系统自动跳停并保持压力，打开压滤机入口蝶阀(125BF)和压滤机流向浓氯化钙清液储槽(4307-TNK-004)的一个蝶阀(150BF)⟹打开烟尘压滤泵(4307-PSO-007/008)出口两个蝶阀(125BF,125BF)、启动烟尘压滤泵(4307-PSO-007/008)，打开入口两个蝶阀(150BF,150BF)⟹料浆通过止推板上的进料孔进去各个滤室，在进料压力不大于0.6MPa下加压过滤，逐渐形成滤饼⟹压滤机滤饼充满后，关闭烟尘压滤泵(4307-PSO-007/008)入口两个蝶阀(150BF,150BF)、停止烟尘压滤泵(4307-PSO-007/008)、关闭烟尘压滤泵(4307-PSO-007/008)出口两个蝶阀(125BF,125BF)和压滤机入口蝶阀(125BF)⟹缓慢打开压缩空气储气罐(4307-CAT-201)一个进出气球阀(50BA)两个出气球阀(50BA,32BA)和出口两个截止阀(32GB,32GB)，在压力不大于0.6MPa下对滤饼进行风干，操作完成后，关闭压缩空气储气罐

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		(4307-CAT-201)一个进出气球阀(50BA)两个出气球阀(50BA,32BA)和出口两个截止阀(32GB,32GB)⟹按住烟尘压滤机(4307-FIL-001)操作面板上的"松开"按钮，压板退回到位后会自动停止，同时拉板钩前进进行往复拉板卸料⟹等待通知，打开或关闭铅锌压滤机(4402-FIL-005/006)过来的滤液一个球阀(200BF)。
10	The Calcium Chloride Concentration Post Stops 氯化钙浓缩岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping; b. Stop in reverse order of driving order , Each equipment needs to stop: Close the two butterfly valves (150BF, 150BF) at the dust filter pump (4307-PSO-007/008), stop the dust filter pump (4307-PSO-007/008), Close the two butterfly valves (4307-PSO-007/008) (125BF, 125BF) and the butterfly valve of filter press (125BF), Press the "Stop" button of the soot filter press (4307-FIL-001), Close the compressed air reservoir (4307-CAT-201) an inlet and outlet air ball valve (50BA) two outlet air ball valves (50BA, 32BA) and the outlet two stop valves (32GB, 32GB)⟹Close the two butterfly valves (500BF, 500BF) of the evaporation circulation pump (4307-PSO-005/006) and stop the evaporation circulation pump (4307-PSO-005/006)⟹Close the two butterfly valves (4307-PSO-001/002), stop the dilute calcium chloride solution pump (4307-PSO-001/002), Close the two butterfly valves (100BF,100BF) (4307-PSO-001/002)⟹Close the draft fan (4307-FAN-001) inlet butterfly valve (2200BF), stop the draft fan (4307-FAN-001) and close the draft fan (4307-FAN-001) outlet butterfly valve (2200BF)⟹Clean the site for hygiene. 8. 正常停车 a. 停车时应待机内物料排净后再停车； b. 按照开车顺序逆序顺序停车，各设备都需要延时停止： 关闭烟尘压滤泵(4307-PSO-007/008)入口两个

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		蝶阀(150BF,150BF)，停止烟尘压滤泵(4307-PSO-007/008)，关闭烟尘压滤泵(4307-PSO-007/008)出口两个蝶阀(125BF,125BF)和压滤机入口蝶阀(125BF)；按下烟尘压滤机(4307-FIL-001)“停止”按钮，关闭压缩空气储气罐(4307-CAT-201)一个进出气球阀(50BA)两个出气球阀(50BA,32BA)和出口两个截止阀(32GB,32GB)关闭电源⟹关闭蒸发循环泵(4307-PSO-005/006)入口两个蝶阀(500BF,500BF)，停止蒸发循环泵(4307-PSO-005/006)⟹关闭稀氯化钙溶液泵(4307-PSO-001/002)入口两个蝶阀(150BF,150BF)，停止稀氯化钙溶液泵(4307-PSO-001/002)，关闭稀氯化钙溶液泵(4307-PSO-001/002)出口两个蝶阀(100BF,100BF)⟹关闭引风机(4307-FAN-001)进口蝶阀(2200BF)，停止引风机(4307-FAN-001)，关闭引风机(4307-FAN-001)出口蝶阀(2200BF)⟹打扫现场卫生。 9. Emergency Stop a. When an emergency occurs, Close smoke pressure filter pump (4307-PSO-007/008), evaporation circulation pump (4307-PSO-005/006), Diline calcium chloride solution pump (4307-PSO-001/002), soot filter press (4307-FIL-001), and draft fan(4307-FAN-001). After all stop the operation, Check whether the start / stop button of each device is in the stop state, Close smoke pressure filter pump (4307-PSO-007/008), evaporation circulation pump (4307-PSO-005/006), Diline calcium chloride solution pump (4307-PSO-001/002) , soot filter press (4307-FIL-001) corresponding valve and draft fan(4307-FAN-001), at the same time, the upper level of equipment all need to stop; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, the operation shall be restarted in the driving sequence;

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№	Stage	Key Points
		c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，关闭烟尘压滤泵(4307-PSO-007/008)、蒸发循环泵(4307-PSO-005/006)、稀氯化钙溶液泵(4307-PSO-001/002)、烟尘压滤机(4307-FIL-001)和引风机(4307-FAN-001)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭烟尘压滤泵(4307-PSO-007/008)、蒸发循环泵(4307-PSO-005/006)、稀氯化钙溶液泵(4307-PSO-001/002)、烟尘压滤机(4307-FIL-001)和引风机(4307-FAN-001)相对应的阀门，同时，上一级设备均需停车； b. 事故处理后，设备恢复正常，如仍需要进行生产，按照开车顺序重新启动操作； c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作
11	Permits &Isolation Sheets 许可证和隔离表	1. Only authorised isolators can sign off and cancel permits and registration sheets. 2. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. Calcium Chloride Concentration Post 氯化钙浓缩岗位 Questionnaire

Declaration of Understanding

I (_____ Learner _____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____ Assessor _____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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9. Appendix

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Standard Operating Procedure
标准操作程序

Calcium chloride concentration posts in the chloride plant
CP 氯化钙浓缩岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
1					
2					
3					

Approvals 批准

Document Owner 程序持有人	Song Yuhua 宋裕华	Date: 日期: 2024.4.24	Signature 签字	
Processing Superintendent 氯化高级副经理	Cao Guibin 曹桂宾	Date: 日期: 2024.4.24	Signature 签字	
Processing Senior Manager 氯化高级经理	Li Jianjun 李建军	Date: 日期: 2024.04.24	Signature 签字	
Safety Department Manager 安全部经理	Airlangga Putra R.J	Date: 日期: 24.04.24	Signature 签字	
Deputy General Manager 副总经理	Li Xinhua 李新华	Date: 日期 2024.4.24	Signature 签字	
General Manager 总经理	Clark Wu 吴迎林	Date: 日期: 2024.4.24	Signature 签字	